SALON MANAGEMENT SYSTEM

Payment Transactions System

Main Project	Salon Management System
Individual project **BY;** **AKAMUMPA MARTIN**	Payment Transactions

1. Database Management System Selection

Main Project

Main Project	Salon Management System

Individual Project	Payment Transactions System

1b. Possible Database Systems

The following DBMS options were considered for this project:

DBMS
Customer database
Appointment Database
Services Database
Payments Database
Transactions Database

Payment and Transactions System

Reasons for selection:

- It tracks all financial transactions in the salon
- It ensures accurate recording of income

- It reduces errors in manual money handling
- It improves accountability and transparency
- It supports different payment methods such as cash and mobile money

2. Business Case

2a. Application Domain

The Payment Transactions System manages all financial operations within the salon. It serves as the core module for recording, tracking, and reporting on customer payments.

The system supports the following core functions:

- Recording customer payments for services rendered
- Tracking which services were paid for in each transaction
- Managing multiple payment methods: cash, mobile money, and card
- Generating receipts for customers and records for management
- Monitoring daily, weekly, and monthly income

System Users:

User Role	Responsibilities
Cashier	Records payments, issues receipts, and processes transa ctions
Salon Manager	Monitors income, views reports, and manages service pri cing
Accountant	Reviews financial records, reconciles transactions, prepar

User Role	Responsibilities
	es reports

2b. Benefits

Tangible Benefits

- Accurate and complete financial records for every transaction
- Faster payment processing, reducing customer wait times
- Reduced manual errors compared to paper-based record keeping
- Easy generation of daily, weekly, and monthly financial reports

Intangible Benefits

- Better management decision-making based on accurate data
- Improved customer trust through professional receipts and records
- Enhanced professional business image

2c. Cost-Benefit Analysis

Cost Item	Details
Computer Hardware	Tangible — physical equipment needed to run the system
Internet Connectivity	Tangible — required for mobile money payment processi ng
Setup & Configuration Tim e	Tangible — initial system installation and testing

Cost Item	Details
Staff Training	Intangible — time investment for staff to learn the system
Adaptation Period	Intangible — temporary productivity dip during transition

Conclusion: Benefits outweigh costs. The system is feasible and recommended for implementation.

2d. User Issues

- Staff may require training to use the system effectively
- Some employees may initially resist switching from manual paper records
- Basic computer literacy is required for all system users

2e. Risk Assessment

Risk	Category & Mitigation
System Crashes	Technical — Regular backups and UPS power backup recommended
Data Loss	Technical — Daily automated database backups to external storage
Staff Non-Compliance	Organizational — Management oversight and training programs
Incorrect Data Entry	Operational — Validation rules and supervisor review procedures

2f. Limitations

- System depends on stable electricity supply to operate
- Mobile money payment processing requires active internet connection
- This is a focused payment module, not a full accounting software suite
- Limited to single-branch operations in the current design

3. Database Design — Logical Design

3a. Entities Overview

Entity	Purpose
Customer	Stores salon customer personal information and contact details
Service	Stores the types of services offered and their prices
Appointment	Links a customer to a service at a specific date and time
Payment	Records the payment made for an appointment

3b. Entities and Attributes

Customer

Attribute	Description
customer_id	Primary key, auto-incremented unique identifie r
name	Full name of the customer
phone	Customer contact phone number
email	Customer email address

Service

Attribute	Description
service_id	Primary key, auto-incremented unique identifie

Attribute	Description
	r
service_name	Name of the service offered (e.g., Haircut, Man icure)
price	Cost of the service in local currency

Appointment

Attribute	Description
appointment_id	Primary key, auto-incremented unique identifie r
customer_id	Foreign key referencing Customer table
service_id	Foreign key referencing Service table
date	Date and time of the appointment

Payment

Attribute	Description
payment_id	Primary key, auto-incremented unique identifie r
appointment_id	Foreign key referencing Appointment table
amount	Total amount paid for the appointment
payment_method	Method used: Cash, Mobile Money, or Card
payment_date	Date and time the payment was recorded

Attribute	Description
status	Payment status: Completed, Pending, or Canc elled

3c. Relationships

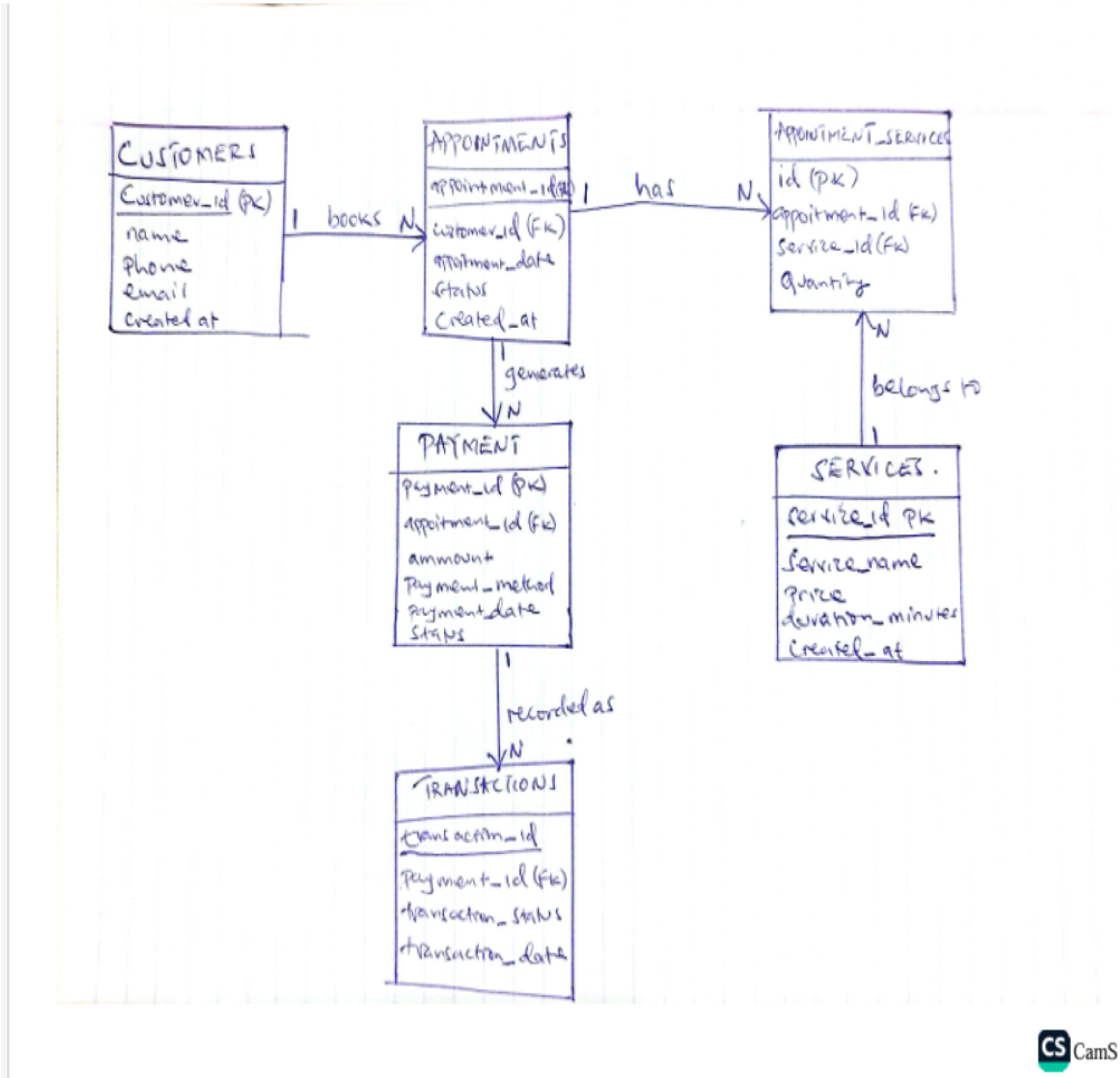
- Customer to Appointment: One customer can have many appointments (1:M)
- Service to Appointment: One service can appear in many appointments (1:M)
- Appointment to Payment: One appointment has one or more payments (1:1 or 1:M)

3d. Multiplicities

Relationship	Multiplicity & Explanation
Customer : Appointment	One-to-Many (1:M) — A customer books multiple appoint ments over time
Service : Appointment	One-to-Many (1:M) — A service can be booked in many ap pointments
Appointment : Payment	One-to-One (1:1) — Each appointment typically has one p ayment record

3e. Entity Relationship Diagram (ERD) Description

The ERD for this system consists of six main entities connected as follows:



4. Physical Design — Database Schema

4a. Payment and Transactions Tables

Customer table

```
MariaDB [payment_transactions]> SELECT *FROM customers;
+-------------+--------+------------+---------------------------+
| customer_id | name   | phone      | email                     |
+-------------+--------+------------+---------------------------+
|           1 | Martin | 0755454648 | akamumpamartin29@gmail.com |
|           2 | Martha | 0723691432 | marthak@gmail.com         |
|           3 | Peter  | 0702689433 | peter21@gmail.com         |
|           4 |  Kim   | 0723865494 | kimber@gmail.com          |
+-------------+--------+------------+---------------------------+
4 rows in set (0.000 sec)
```

Services Table

```
MariaDB [payment_transactions]> SELECT * FROM services;
+------------+---------------+-----------+
| service_id | service_name  | price     |
+------------+---------------+-----------+
|          1 | Haircut       |  10000.00 |
|          2 | Manicure      |  25000.00 |
|          3 | Massage       | 100000.00 |
|          4 | Hair Coloring |  35000.00 |
+------------+---------------+-----------+
```

Appointment Table

```
MariaDB [payment_transactions]> SELECT * FROM appointments;
+----------------+-------------+------------------+-----------+
| appointment_id | customer_id | appointment_date | status    |
+----------------+-------------+------------------+-----------+
|              1 |           1 | 2026-05-07       | Completed |
|              2 |           2 | 2026-05-09       | Completed |
|              3 |           3 | 2026-05-10       | Pending   |
|              4 |           4 | 2026-05-12       | Completed |
+----------------+-------------+------------------+-----------+
4 rows in set (0.000 sec)
```

Payment Table

```
MariaDB [payment_transactions]> SELECT * FROM payments;
+------------+----------------+-----------+----------------+---------------------+-----------+
| payment_id | appointment_id | amount    | payment_method | payment_date        | status    |
+------------+----------------+-----------+----------------+---------------------+-----------+
|          1 |              1 |  10000.00 | Cash           | 2025-05-05 00:00:00 | Completed |
|          2 |              2 |  25000.00 | Mobile Money   | 2025-05-05 00:00:00 | Completed |
|          3 |              3 | 100000.00 | Card           | 2025-05-05 00:00:00 | Pending   |
|          4 |              4 |  35000.00 | Cash           | 2025-05-05 00:00:00 | Completed |
+------------+----------------+-----------+----------------+---------------------+-----------+
4 rows in set (0.000 sec)
```

5. Payments and Transactions system Queries

Query 1

Total revenue and payment methods

```
MariaDB [payment_transactions]> SELECT SUM(amount) AS total_revenue
    -> FROM payments
    -> WHERE status = 'Completed';
+---------------+
| total_revenue |
+---------------+
|      70000.00 |
+---------------+
1 row in set (0.001 sec)

MariaDB [payment_transactions]> SELECT payment_method, SUM(amount) AS total
    -> FROM payments
    -> GROUP BY payment_method;
+----------------+-----------+
| payment_method | total     |
+----------------+-----------+
| Card           | 100000.00 |
| Cash           |  45000.00 |
| Mobile Money   |  25000.00 |
+----------------+-----------+
3 rows in set (0.001 sec)

MariaDB [payment_transactions]> UPDATE payments
    -> SET status = 'Completed'
    -> WHERE payment_id = 3;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Query 2

Customer Payment History

Lists all payments with customer names and appointment details:

```
MariaDB [payment_transactions]> SELECT *
    -> FROM payments
    -> WHERE status = 'Completed';
+------------+----------------+----------+----------------+---------------------+-----------+
| payment_id | appointment_id | amount   | payment_method | payment_date        | status    |
+------------+----------------+----------+----------------+---------------------+-----------+
|          1 |              1 | 10000.00 | Cash           | 2025-05-05 00:00:00 | Completed |
|          2 |              2 | 25000.00 | Mobile Money   | 2025-05-05 00:00:00 | Completed |
|          4 |              4 | 35000.00 | Cash           | 2025-05-05 00:00:00 | Completed |
+------------+----------------+----------+----------------+---------------------+-----------+
3 rows in set (0.004 sec)

MariaDB [payment_transactions]> SELECT c.name, p.amount, p.payment_method
    -> FROM customers c
    -> JOIN appointments a ON c.customer_id = a.customer_id
    -> JOIN payments p ON a.appointment_id = p.appointment_id;
+--------+-----------+----------------+
| name   | amount    | payment_method |
+--------+-----------+----------------+
| Martin |  10000.00 | Cash           |
| Martha |  25000.00 | Mobile Money   |
| Peter  | 100000.00 | Card           |
|  Kim   |  35000.00 | Cash           |
+--------+-----------+----------------+
4 rows in set (0.005 sec)
```

Query 3

ALL Entities combined (Joined)

```
MariaDB [payment_transactions]> SELECT
    ->     c.name,
    ->     s.service_name,
    ->     p.amount,
    ->     p.payment_method,
    ->     t.transaction_status,
    ->     p.payment_date
    -> FROM customers c
    -> JOIN appointments a ON c.customer_id = a.customer_id
    -> JOIN appointment_services aps ON a.appointment_id = aps.appointment_id
    -> JOIN services s ON aps.service_id = s.service_id
    -> JOIN payments p ON a.appointment_id = p.appointment_id
    -> JOIN transactions t ON p.payment_id = t.payment_id;
+--------+---------------+-----------+----------------+--------------------+---------------------+
| name   | service_name  | amount    | payment_method | transaction_status | payment_date        |
+--------+---------------+-----------+----------------+--------------------+---------------------+
| Martin | Haircut       |  10000.00 | Cash           | Success            | 2025-05-05 00:00:00 |
| Martha | Manicure      |  25000.00 | Mobile Money   | Success            | 2025-05-05 00:00:00 |
| Peter  | Massage       | 100000.00 | Card           | Pending            | 2025-05-05 00:00:00 |
|  Kim   | Hair Coloring |  35000.00 | Cash           | Success            | 2025-05-05 00:00:00 |
+--------+---------------+-----------+----------------+--------------------+---------------------+
4 rows in set (0.003 sec)
```

6. Summary

The Payment Transactions System is a well-structured, feasible module within the broader Salon Management System. It addresses all core business requirements: accurate financial records, support for multiple payment methods, customer history tracking, and management reporting. It is cost-effective, scalable, and ready for implementation using standard open-source tools.

End of Report

Salon Management System — Payment Transactions System